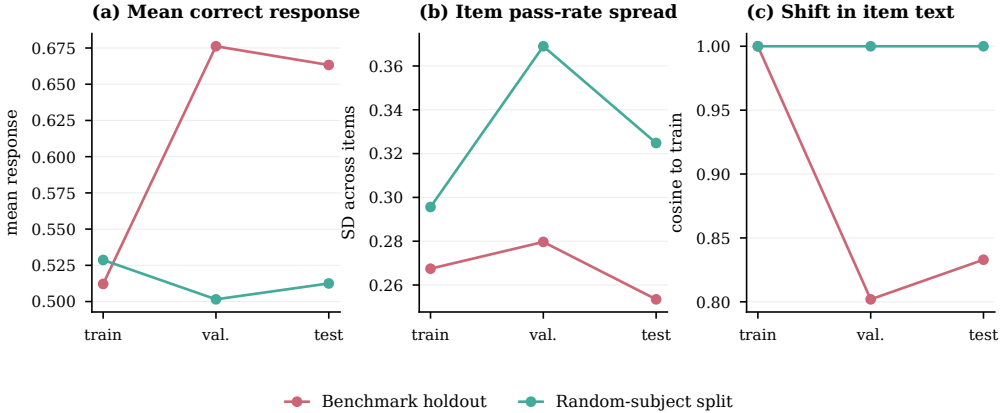
A random-subject split is not a test of benchmark transfer



(d) Retained responses after filtering

Split	Role	Responses	Tasks	Items
Benchmark holdout	train	5,359,699	6,013	109,987
	validation	812,782	390	3,451
	test	166,450	78	9,597
Random-subject split	train	2,620,414	4,073	45,047
	validation	552,360	939	45,047
	test	1,174,388	1,252	56,308

Figure. The same filtered responses split two ways. The benchmark holdout keeps whole benchmarks out of training, so validation and test items are new text (c) with a different mean response (a). The random-subject split holds out subjects but keeps every benchmark in training, so its held-out items match training exactly (cosine 1.0): it measures new subjects, not new benchmarks, and is not a transfer baseline. The two splits use different frozen embedding models, so compare cosines within a split only. Sources: docs/benchmark-holdout-split-distribution-summary.csv and docs/non-benchmark-holdout-split-distribution-summary.csv.

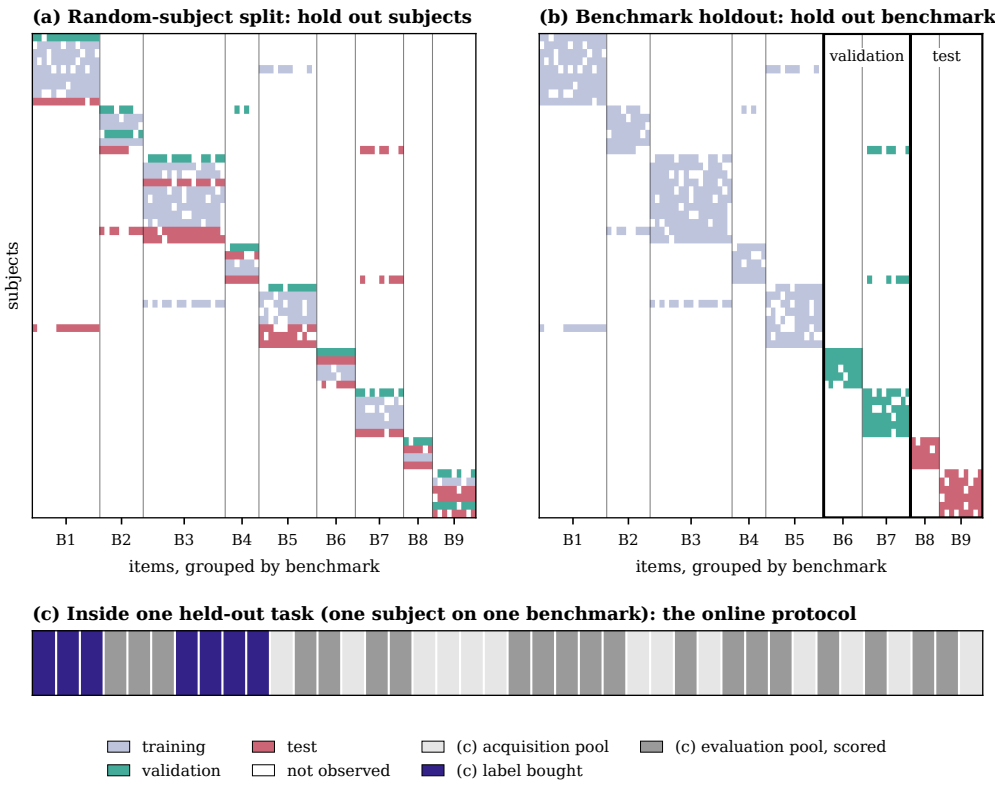
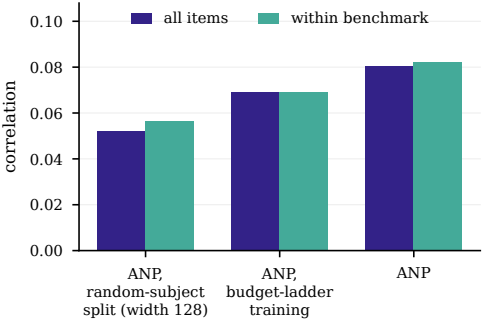


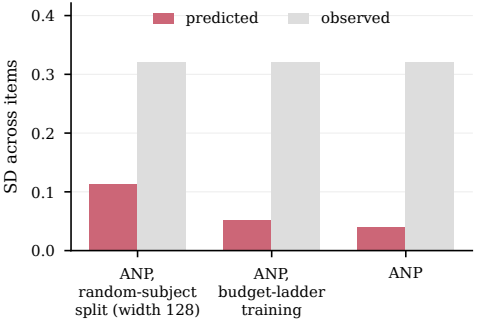
Figure. Schematic of the subject x item response matrix (shape, not scale): items are grouped into benchmarks of unequal size, and each subject answers mostly one benchmark, so the observed cells form blocks. (a) The random-subject split holds out subjects inside every benchmark, so every benchmark is seen in training. (b) The benchmark holdout holds out whole benchmarks (outlined): validation and test items come from benchmarks never seen in training; ours uses 11 training, 3 validation and 3 test benchmarks. (c) Within each held-out task the items are split once, 50/50, into an acquisition pool, from which up to 31 labels are bought one at a time in stream order, and a fixed evaluation pool that is never revealed and is scored at every budget.

The ANPs barely predict which new items are hard

(a) Predicted vs observed item pass rate



(b) Predictions are compressed



(c) Two protocols, test split

Model	Corr.	Within	Pred. SD	Obs. SD
ANP, random-subject split (width 128, Qwen3-8B embeddings)	0.339	0.277	0.109	0.325
ANP (benchmark holdout, BGE-M3 embeddings)	0.080	0.082	0.040	0.321

Figure. Item-difficulty transfer on 297 held-out items, at zero labels. (a) Correlation between each item's predicted and observed pass rate, over all items and within each benchmark. (b) The spread of the predictions is a fraction of the observed spread: the models predict nearly the same probability for every item. (c) The same contrast for a random-subject-split model scored on its own test split. The two rows differ in split, encoder and evaluation set, so this is evidence of a transfer bottleneck, not an architecture ablation. Sources: docs/np-online-passrate-figure11-summary.csv and docs/non-benchmark-anp128-qwen-item-passrate-summary.csv.

The held-out benchmarks differ in difficulty more than the subjects do

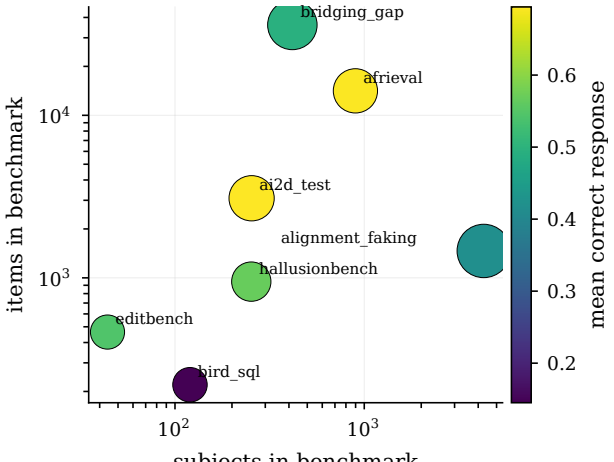
Role	Benchmark	Items	Subjects	Mean	SD	10%	90%
Train	appworld	462	18	0.28	0.24	0.05	0.63
	alignment_faking	1,464	4,278	0.41	0.29	0.00	0.78
	cruxeval	1,245	26	0.42	0.13	0.28	0.54
	bridging_gap	35,984	417	0.46	0.14	0.33	0.69
	helm_afr	28,662	15	0.52	0.08	0.46	0.64
	hallusionbench	948	253	0.56	0.11	0.43	0.69
	harmbench	801	29	0.59	0.27	0.22	0.87
	afrimedqa	7,900	31	0.63	0.13	0.46	0.78
	brace	4,987	28	0.66	0.14	0.40	0.83
	afrieval	14,141	898	0.68	0.11	0.59	0.79
	agi_elo	13,393	20	0.70	0.21	0.36	0.87
Validation	bird_sql	220	120	0.15	0.18	0.00	0.40
	financebench	143	16	0.50	0.32	0.14	0.89
	ai2d_test	3,088	254	0.69	0.15	0.50	0.86
Test	editbench	464	44	0.54	0.10	0.39	0.65
	cmmlu	8,994	16	0.68	0.16	0.47	0.84
	algotune	139	18	0.69	0.13	0.52	0.83

	All subjects in the role	Subjects	Mean	SD	10%	90%
Train	mean accuracy per subject	6,013	0.47	0.27	0.02	0.78
Validation	mean accuracy per subject	390	0.52	0.30	0.00	0.84
Test	mean accuracy per subject	78	0.61	0.14	0.44	0.79

Table. Difficulty of every benchmark in the benchmark-holdout split, as subject accuracy on it: each subject's accuracy on the benchmark, then the mean, standard deviation and 10th/90th percentiles over subjects (lower means harder). The lower block pools each role's subjects. Validation holds the hardest benchmark (bird_sql, 0.15) and test holds three mid-to-easy ones (0.54-0.69), so test subjects look stronger than training subjects (0.61 against 0.47). Much of that gap is the training side: alignment_faking contributes 4,278 of the 6,013 training subjects (71%), many near zero, which pulls the per-subject training average down. With three benchmarks per held-out role, which benchmarks are held out moves these averages substantially.

What is analysed in this appendix

Two different objects are kept separate: observed binary responses and frozen item-text embeddings.



Response graph

A bipartite edge joins a subject to an item only when that binary response was observed. Missing responses are not imputed.

Qwen item graph

Each filtered item connects to its 20 nearest 1024-d Qwen vectors by cosine similarity. It uses question text only.

Why separate them

Response structure needs labels and is training-only. Qwen edges can be built for a new item before its label is known.

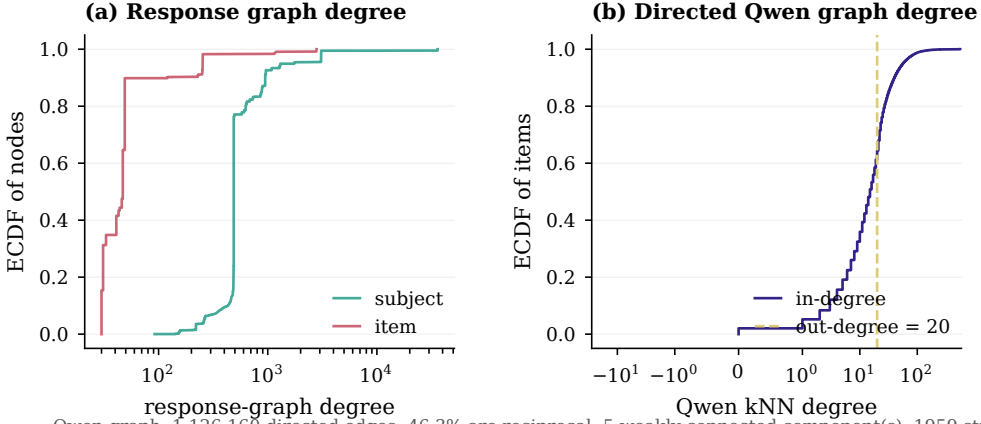
Filtering: 16,278,271 raw rows to 5,150,869; item support >= 30, subject-task responses >= 80.

How the two graphs are constructed

The response graph and the Qwen kNN graph have different edges, degree distributions, and test-time status.



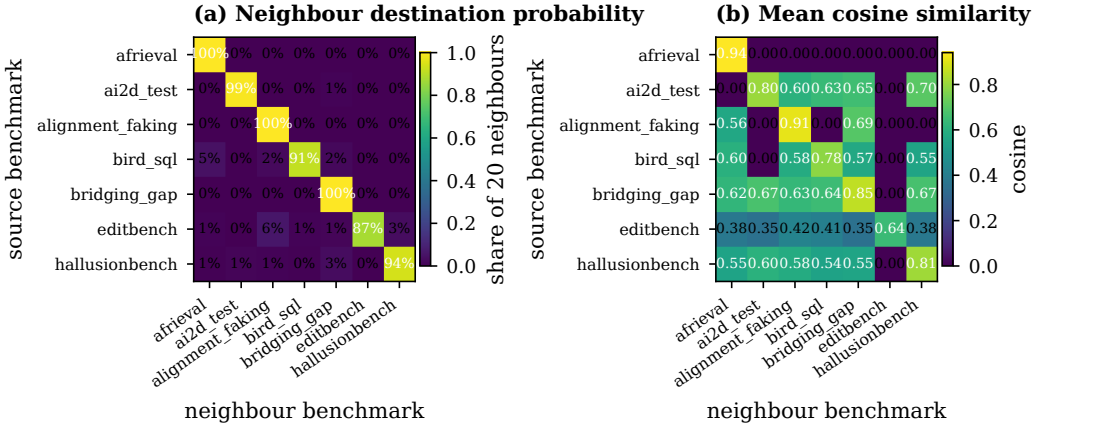
Direction comes from the source item's own Top-K list. It need not be mutual: $i \to j$ can exist while $j \not\to i$ does not.

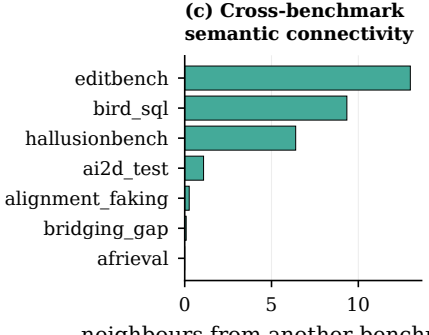


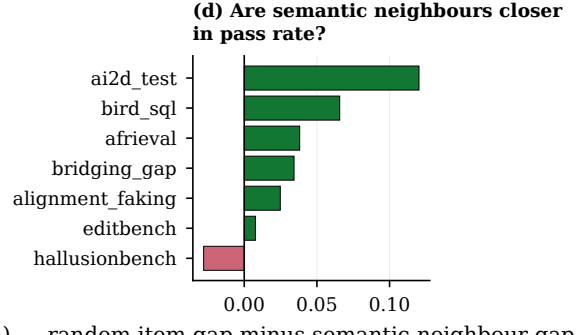
Qwen graph: 1,126,160 directed edges; 46.3% are reciprocal; 5 weakly connected component(s), 1959 strongly connected component(s).

Qwen3 item similarity forms a content-only graph

Each row contains the 20 nearest text-embedding neighbours of items from that benchmark. It is directed because each item chooses its own neighbours.



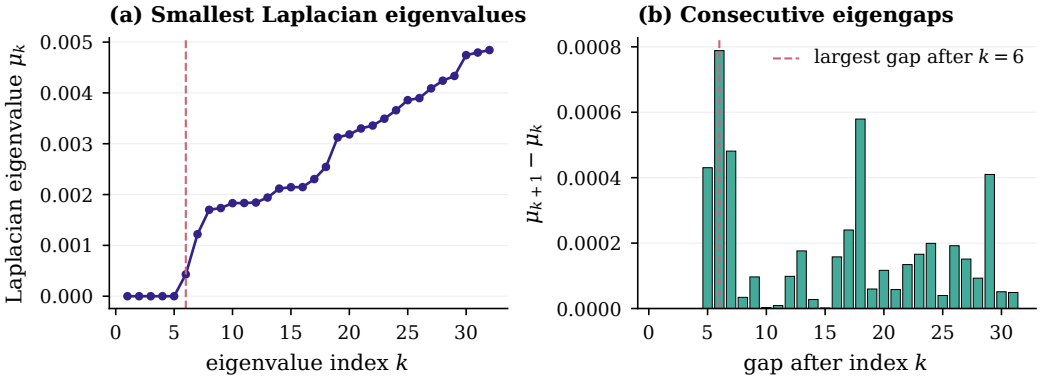




Panel (d): positive means Qwen neighbours have more similar observed pass rates than random items, after matching source and target benchmarks. It is descriptive, not a test-time score.

Laplacian spectrum of the symmetrised Qwen kNN graph

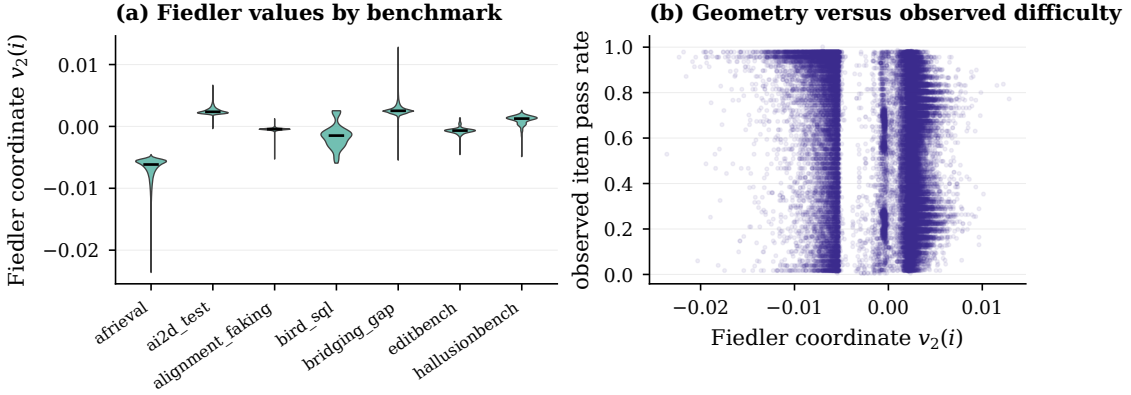
Cosine chooses each item's 20 neighbours; W is then the binary symmetrised kNN adjacency. $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$. Near-zero values count disconnected pieces; a later eigengap suggests a coarse cluster count.



There are 5 weak components, so the first 5 eigenvalues should be near zero. The largest displayed gap is after k=6. This is a heuristic, not proof that k clusters are stable.

What the Fiedler vector can, and cannot, show

The Fiedler vector is the eigenvector for the second-smallest μ_2. It is computed inside the largest weak component because the global graph has several disconnected components and no unique global Fiedler vector.



Largest weak component: 56,221 items. The sign of v_2 is arbitrary. Separation by benchmark means the graph retains benchmark structure; a smooth relation to pass rate would suggest a difficulty-relevant semantic direction.